

Coupling-gated cortical coherence: a conditional framework for unity and self-representation

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Abstract

Coordinated cortical activity need not describe a single experienced situation. We develop a conditional field account of coherence, compatibility and self-representation. Its demonstrated results establish what coordination leaves undetermined: increasing coupling, decreasing diffusion and a matched common drive can produce identical stationary phase distributions; equal phases permit incompatible local states; and a plasticity control preserves coherence while losing alignment with a specified environmental template. The conditional construction glues compatible spatial measures into one state and makes that state the unique fixed point of a contracting reconstruction map. Lean 4 checks the composition under eight connecting hypotheses and additional structures, with compatibility and accuracy at the glued state supplied as premises. We propose the glued state as a formal correlate of conscious unity and its accurate internal reconstruction as a correlate of a minimal self. The physical hypothesis is that changing extracellular geometry increases field-mediated coupling during awakening. A noisy mean-field phase model gives a square-root stationary onset, but finite ramps delay the response and the distributional

equivalence requires measurements that distinguish recovery mechanisms. Testing the account further requires identifying neural content and self-encoding independently, then determining whether coordination maintains their agreement and whether these relations matter for unified experience.

1 Introduction

How could activity distributed across cortex support one experience? A field account proposes that endogenous extracellular electromagnetic fields partly coordinate spatially separated neural processes (McFadden, 2002, 2020; Pockett, 2012). Ephaptic effects provide a possible interaction mechanism (Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015). Yet even successful coordination leaves a question: what makes the coordinated activity a description of one experienced situation, with a perspective on that situation?

We propose an account with three requirements. *Coherence* concerns whether coupling sustains collective activity. *Compatibility* concerns whether local descriptions agree where their domains overlap. *Self-representation* concerns whether the resulting global state supports an accurate model of itself. The latter two express commitments about consciousness whose motivation and possible dissociations we develop below.

Consider the experience of reaching for a cup. A description of the cup's location and a description of the hand's relation to it can share information about where the cup is while contributing different contents to the episode.

19 Synchronous activity in the two processes would leave any disagreement
20 about that shared quantity unresolved. The formal counterexample isolates
21 this problem using spatial mass profiles at a common phase that disagree
22 on an overlap. We return to the cup example to ask what agreement would
23 mean for perceptual content, and what an internal representation would have
24 to add to place the perceiver in that situation. It is an illustration of the
25 modelling task, not a neural model implemented here.

26 Awakening gives the physical requirement a concrete setting. We hypoth-
27 esize that changing extracellular geometry slowly increases effective coupling
28 through a synchronization threshold. Under stationary branch tracking, a
29 noisy phase model fixes a square-root onset of coherence. The experiment
30 faces two distinct obstacles. Finite ramp speeds delay the response, so a
31 fitted onset exponent depends on the drive. Moreover, successive stationary
32 phase distributions can be identical under increasing coupling, decreasing
33 diffusion or a suitably matched common drive. Recovery must therefore be
34 measured in a way that identifies dynamics and distinguishes causes, rather
35 than inferred from the shape of a coherence trace alone.

36 A plasticity control tests another route to representation. Its squared-
37 drift objective rewards equalizing drift, which in the chosen frequency-cluster
38 model favors connections across the clusters that define the diagnostic tem-
39 plate. Coherence survives while alignment with that template declines. To-
40 gether with the overlap counterexample, this result establishes a limit on
41 what the coordination model explains. Compatible local content needs its

own mechanism and observable.

The contribution has three levels. The separation results establish limits on what coordination explains. The conditional construction gives a unique global state from specified compatible spatial measures and makes it the unique fixed point of a contracting map assumed to reconstruct that state. Lean 4 checks the composition, and toy witnesses establish joint satisfiability of its assumptions. The biological interpretation proposes that the glued state and its internal reconstruction capture conscious unity and a minimal self. This last step requires identifying meaningful neural descriptions and explaining how coordination maintains their agreement. Figure 1 locates the proposed connections, and Section 6 states the physical and thermodynamic assumptions.

2 Coherence: the coupling model

2.1 Dynamics and the stationary threshold

The finite stochastic model is

$$d\theta_i = \left[\omega_i + \sum_j K_{ij}(t) \sin(\theta_j - \theta_i) \right] dt + \sqrt{2D} dW_i, \quad r_N = \left| \frac{1}{N} \sum_i e^{i\theta_i} \right|. \quad (1)$$

Here $D > 0$ is phase diffusion, W_i are independent Wiener processes, and K_{ij} has inverse-time units, with angles dimensionless. The uniform mean-field convention is $K_{ij} = K/N$. Frequency heterogeneity and phase diffusion are

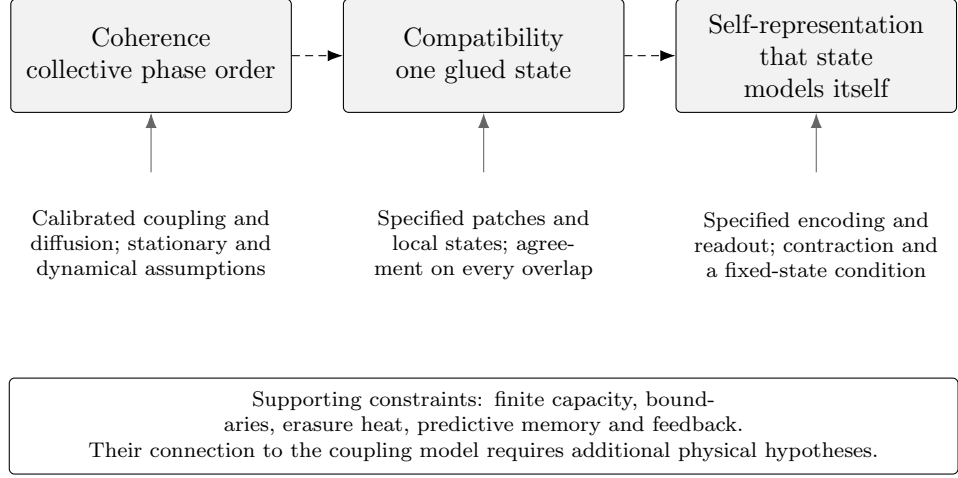


Figure 1: Three requirements of the proposed account. Each requires the inputs shown below it. Dashed arrows mark proposed connections: phase coherence alone supplies neither compatible local descriptions nor an accurate self-model. Thermodynamics constrains a candidate implementation through separate assumptions. This is the explanatory structure of the article; Table 1 records the eight implications consumed by each branch of the formal composition.

60 separate parameters.

61 For identical frequencies in their rotating frame, the stationary ansatz is

$$\rho(\theta) = \frac{e^{a \cos(\theta - \psi)}}{2\pi I_0(a)}, \quad a = \frac{Kr}{D}, \quad r = R(a) := \frac{I_1(a)}{I_0(a)}, \quad (2)$$

62 where I_n are modified Bessel functions and ψ is the collective phase. Lean
 63 verifies normalization, the circular order parameter and self-consistency for
 64 this density. At $K \leq K_c = 2D$, the only nonnegative self-consistent order
 65 parameter is zero. At $K > K_c$, exactly one positive solution coexists with

66 zero; it emerges continuously and increases strictly with K . Near threshold,

$$\frac{r^2}{K - K_c} \rightarrow \frac{1}{D}, \quad r \sim \sqrt{\frac{K - K_c}{D}}. \quad (3)$$

67 These results characterize the stationary branch (Fig. 2). Connecting it
 68 to finite trajectories requires a dynamical mean-field limit and dynamical
 69 selection, neither proved here. Propagation of chaos concerns the first of
 70 these steps (Sznitman, 1991); exchange symmetry alone does not establish
 71 independence, especially when realizations have an unpinned collective phase.

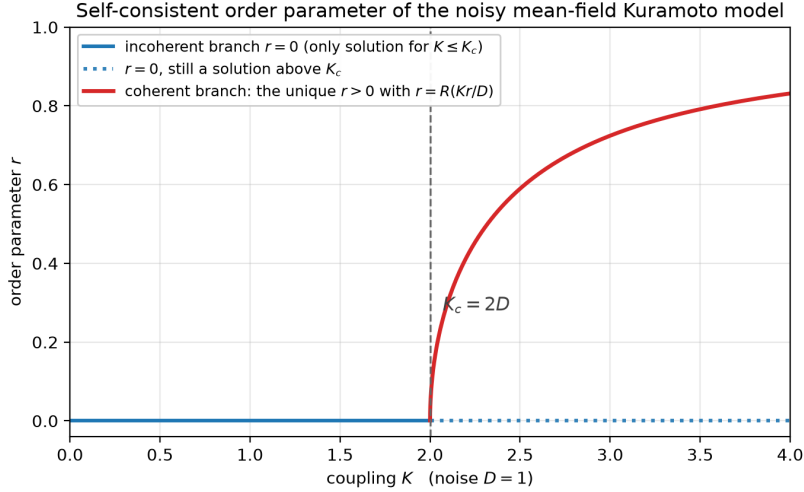


Figure 2: Stationary self-consistency for identical frequencies and phase diffusion $D = 1$. The positive coherent branch appears continuously above $K_c = 2D$ and increases with coupling. Zero remains a solution above threshold. The curve is computed from the integral expressions used in the formalization.

72 A spatial version replaces the finite sum by an integral against a kernel
 73 $K(x, y)$. The substrate measure has total mass one, making the mean-field

74 strength a kernel average. Applying the scalar threshold to a heterogeneous
 75 or signed kernel requires a justified reduction to that mean-field description.

76 **2.2 Physical identification and calibration**

77 The cortical hypothesis identifies part of the effective coupling with endoge-
 78 nous extracellular electromagnetic interactions. Fields can modulate spike
 79 timing (Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015), and field
 80 models have been proposed for coordinating distributed neural information
 81 (Pinotsis & Miller, 2022; McFadden, 2020). The quantitative question is
 82 whether that interaction supplies sufficient coupling relative to diffusion.
 83 Changes in phase organization under propofol (Bastos et al., 2021) moti-
 84 vate studying this regime, but do not measure the coupling itself.

85 The physical field predicate requires a jointly continuous kernel, a nor-
 86 malized substrate, a mean-field strength equal to the spatial average, and no
 87 single site providing more than half that strength. Regular mesh refinement
 88 establishes convergence of a scalar coupling-energy quadrature; constructing
 89 the kernel and relating it to trajectories are separate modelling tasks. The
 90 formal witnesses supply toy instances of the predicate.

91 Calibration is essential because a spike-time shift is not a coupling rate.
 92 An estimate proportional to $N\Delta tf$, using population size, timing shift and
 93 oscillation frequency, is dimensionless. A conversion of the form

$$K_{\text{eff}} = \gamma N s E f \tag{4}$$

94 requires a rate γ , where s is the timing shift per unit field and E the field
 95 amplitude. This rate specifies how induced timing shifts persist and accumu-
 96 late into phase drift. Supplemental Material explains why the other factors
 97 cannot determine it. The biological threshold test requires independent es-
 98 timates of the interaction kernel, its aggregation, and phase diffusion in the
 99 same cortical regime.

100 **3 Testing coherence through recovery after** 101 **awakening**

102 The proposed control variable is extracellular geometry. Sleep-associated
 103 changes in extracellular space and clearance (Xie et al., 2013) motivate a
 104 coupling parameter that evolves more slowly than phase dynamics. The
 105 extracellular volume fraction is approximately $\alpha = 0.2$ at diffusion tortuosity
 106 $\lambda \approx 1.6$ (Syková & Nicholson, 2008); the cited sleep study reports a rise
 107 in α from roughly 0.14 to 0.23 (Xie et al., 2013). Below 10 kHz cortical
 108 tissue conducts approximately resistively at $0.3\text{--}0.6\text{ S m}^{-1}$ (Barbour, 2017).
 109 In an effective-medium description at fixed transmembrane current density,
 110 a larger extracellular volume fraction raises conductivity and reduces field
 111 amplitude.

112 This gives the coupling-gated hypothesis a directional prediction: contrac-
 113 tion of extracellular space on awakening raises E and, with the other factors
 114 in Eq. (4) held fixed, raises effective coupling. The proposed geometry–

115 coupling relation and its role in human recovery both require measurement.
 116 The effective-medium argument supplies a sign under stated conditions; cal-
 117 ibration supplies the magnitude and the mapping to the scalar model.

118 If $K(t)$ rises through K_c and the phase distribution tracks the coherent
 119 stationary branch, Eq. (3) gives

$$r(t) \sim \sqrt{\frac{\dot{K}(t_0)}{D}} (t - t_0)^{1/2}, \quad t > t_0, \quad \dot{K}(t_0) > 0. \quad (5)$$

120 The crossing time satisfies $K(t_0) = K_c$. It is a prediction of latency only when
 121 coupling is measured independently of coherence. The subsequent shape
 122 and saturation depend on the later coupling trajectory. Critical slowing
 123 down restricts branch tracking near threshold, while finite populations have
 124 fluctuating order below it. A measurable onset therefore needs an explicit
 125 criterion above that fluctuation floor.

126 **3.1 What a recovery distribution can identify**

127 Equation (2) gives a concentration–coherence relation with no fitted shape
 128 parameter. Testing it requires separate estimators: r from circular resultant
 129 length and a from density shape. Estimating a by inverting R from the
 130 measured r makes agreement automatic. Even a valid agreement tests a
 131 statistical description shared by the von Mises family; it does not determine
 132 the coupling medium.

133 There is a stronger limitation. Successive equilibrated phase distributions

134 cannot, by themselves, distinguish three different recovery mechanisms. Let
 135 $q(t)$ be a prescribed ratio crossing 2, let $r(q)$ denote its stationary branch,
 136 and choose a fixed rate $D_0 > 0$. Increasing coupling with $K(t) = D_0 q(t)$
 137 and $D = D_0$ gives the same stationary density as fixed coupling K_0 with de-
 138 creasing diffusion $D(t) = K_0/q(t)$. Both have concentration $a = q(t)r(q(t))$.
 139 Uncoupled oscillators under a common prescribed drift $-h(t) \sin(\theta - \psi)$, with
 140 $h(t) = D_0 q(t)r(q(t))$ and $D = D_0$, have that same density. Their distribu-
 141 tions coincide window by window, even though their physical causes differ.
 142 Supplemental Material supplies the construction and its numerical feasibility
 143 check.

144 The missing observable is the physical clock. Resolved phase increments
 145 contain rate information: in the ideal continuous-path model, quadratic vari-
 146 ation accumulates at rate $2D$, while conditional increment means constrain
 147 drift. This can separate coupling and diffusion under the specified observa-
 148 tion model. Distinguishing endogenous coupling from the matched common
 149 drive additionally requires a measured drive or a controlled perturbation, be-
 150 cause their mean-field drifts can coincide. Identifying the coupling as electro-
 151 magnetic then requires evidence specific to that interaction. This hierarchy
 152 determines the experiment: distribution, dynamics and physical mechanism
 153 are distinct targets.

154 **3.2 Finite ramps change the observable onset**

155 Finite simulations characterize how the stationary prediction can appear in
156 a recovery measurement. The ramp study uses $N = 2000$, $D = 1$ and 32
157 replicas at four coupling speeds. Escape is the first of three consecutive
158 saved samples at $r \geq 0.2$ after threshold. The supplement gives parameters,
159 estimators and further controls.

160 Faster ramps delay escape (Fig. 3). Only 5/32 replicas escape within the
161 fastest window. The three slower ramps have mean coupling delays 0.1947,
162 0.0684 and 0.0253. Their fitted onset exponents run from 0.898 to 0.417,
163 compared with 0.443 to 0.448 from the same estimator on the exact stationary
164 branch at the same coupling grids. At the fastest speed the ensemble mean
165 never reaches the declared fit window; the returned value 0.964 is a local
166 log-slope of a delayed trajectory.

167 This is delayed passage through a noisy bifurcation (Berglund & Gentz,
168 2002), with a direct consequence for the awakening test: the apparent expo-
169 nent depends on the ramp and on the fitting window. A deviation from the
170 square-root law constrains the adiabatic approximation only where branch
171 tracking is independently justified. The sweep does not bracket the specified
172 deviation-based critical ramp speed.

173 Additional controls test the approximations used by the physical model.
174 Time-step refinement attributes most of the sampled discrepancy from the
175 stationary order to integration bias. Population-size diagnostics show why
176 a shared random collective orientation can preserve unconditional pair cor-

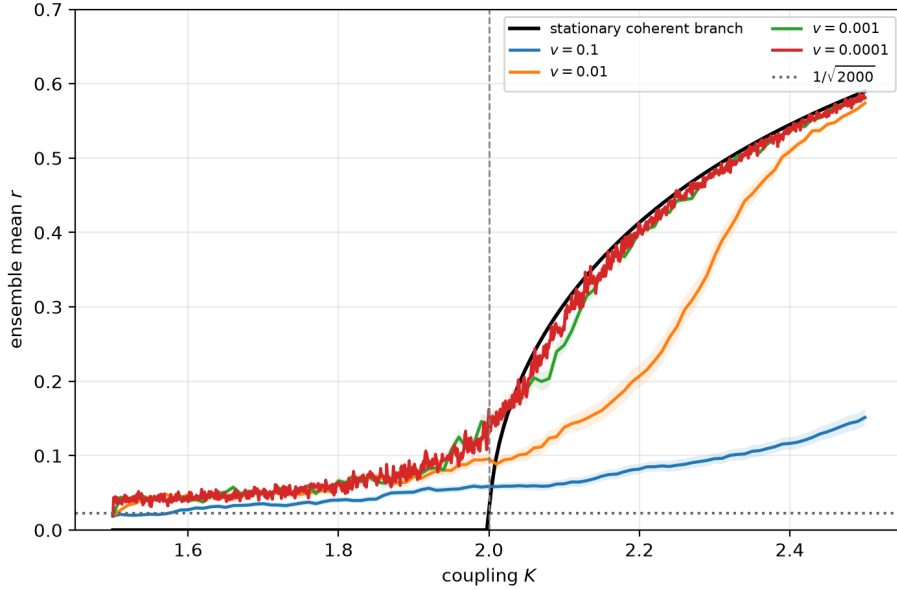


Figure 3: Finite-rate departures from the stationary branch. Ensemble-mean order versus coupling for four ramp speeds v , with one-standard-error bands across replicas. The dashed vertical line is the static threshold and the dotted horizontal line the finite-size reference. Faster ramps delay escape, changing the apparent recovery onset.

177 relation in the coherent regime. Spatial and signed-network controls exhibit
 178 coherence at specified total coupling, with the refined signed-network cross-
 179 ing bracket $K_{\text{eff}}/D \in [1.958, 2.009]$ still containing 2. These controls delimit
 180 finite regimes and numerical resolution; the supplement reports their full
 181 evidence. Calibrating the cortical coupling remains the experimental task.

182 3.3 What the estimators produce on EEG

183 Public propofol-sedation EEG (Bajwa et al., 2025) provides an exploratory
 184 exercise of the concentration and coherence estimators. The implemented

185 analysis pools channels over 100-ms windows, estimating a spatiotemporal
186 phase distribution. Bipolar and common-reference montages produce differ-
187 ent residuals. The bipolar results are compatible with the Bessel relation,
188 but the concentration range is too narrow to distinguish it from its linear
189 approximation at the observed scatter.

190 An informative recording needs a broader concentration range, estima-
191 tor calibration that accounts for dependence, and an observation model for
192 source mixing, referencing and temporal pooling. The sleep-recovery hypoth-
193 esis remains untested by these sedation recordings.

194 **3.4 A protocol for the awakening experiment**

195 The analysis defines six requirements for the physical test.

- 196 **1. Coupling and diffusion in the same units.** Estimate K and D
197 from resolved phase dynamics in the same cortical regime, separating
198 diffusion, frequency heterogeneity and observation noise. A measured
199 drive or controlled perturbation is additionally needed to distinguish
200 endogenous coupling from the matched common-drive alternative.
- 201 **2. Geometry calibrated to coupling.** Establish the relation between
202 an extracellular-geometry observable and effective coupling, with un-
203 certainty. A diffusion-MRI or astrocytic proxy for volume fraction re-
204 quires its own validation; the conductivity argument supplies only the
205 expected sign.

- 206 **3. Adequate phase measurements.** For the proposed spatial estima-
207 tor, use at least 100 sites per estimate, r from resultant length, and
208 a from unweighted regression of log density on the cosine of centered
209 phase. This numerical minimum needs null calibration at the actual
210 sample count, concentration range and dependence. Temporal pooling
211 changes the estimand.
- 212 **4. An independent recovery endpoint.** Measure behavioral recovery
213 separately from the phase trace.
- 214 **5. A predeclared onset criterion.** Fix a coherence threshold above the
215 finite-size floor and account for finite ramp speed before interpreting
216 an onset exponent.
- 217 **6. Matched model comparison.** Compare coupling-gated recovery
218 with diffusion change, common drive and alternative slow-recovery
219 models under the same observation and filtering assumptions. Station-
220 ary distributions alone cannot resolve the equivalence in Section 3.1.

221 Failure of the concentration–coherence relation over an informative range
222 would constrain the phase-density model. Failure of the calibrated geometry–
223 coupling association would constrain the proposed physical gate. A com-
224 peting dynamical explanation that better predicts independently measured
225 recovery would count against its role in awakening. These tests address the
226 first requirement of the framework. Even a favorable outcome leaves the
227 question of what the coordinated activity represents.

228 4 Compatibility: from coordinated activity 229 to a global description

230 4.1 Why common timing is insufficient

231 Consider overlapping patches U and V , each carrying a local description.
232 Compatibility requires that restricting either description to $U \cap V$ gives the
233 same object. Agreement concerns the shared domain; descriptions on distinct
234 parts can remain different. For the finite spatial measures used here, equal
235 overlap restrictions give a unique measure on the covered domain. This is
236 the gluing property formalized by a sheaf.

237 In the cup example, suppose a visual description locates the cup on a
238 table and a reaching description relates the hand to the cup. To compare
239 them, a model must express their shared quantity, such as cup location, in
240 a common coordinate system. Restriction would retain that quantity while
241 dropping details peculiar to each description. Agreement on it permits the
242 visual description to contribute surface appearance and the reaching descrip-
243 tion to contribute hand configuration without duplicating the whole situa-
244 tion. Disagreement after coordinate conversion would make the two estimates
245 incompatible at that level of description, however closely timed their activity.

246 The formal counterexample makes the additional requirement visible. Ev-
247 ery patch has the same oscillator phase, yet two local mass profiles assign
248 different densities to a shared site. They cannot be restrictions of a single

249 global measure. Phase order therefore cannot establish compatibility. Con-
250 tinuity of a physical field does not repair disagreement in the descriptions
251 associated with it.

252 For a theory of experience, the attraction of gluing is this contribution
253 of different local contents to one description. The cup example does not yet
254 instantiate the theorem: receptive territories and content domains must be
255 related to the spatial patches, and their descriptions must have the required
256 extension property. In particular, a decoder assigning probability laws to per-
257 ceptual contents needs a further extension model, because matching overlap
258 marginals need not determine a unique global joint distribution (Abramsky &
259 Brandenburger, 2011). The spatial-measure construction provides a precise
260 case of gluing; a content model must justify its own existence and uniqueness
261 claims.

262 **4.2 Coherent adaptation can lose environmental align-** 263 **ment**

264 An adaptive coupling rule might appear to supply that missing representa-
265 tional structure. The squared-drift control shows why the objective mat-
266 ters. Symmetric coupling conserves mean drift: pairwise sine terms cancel
267 when summed over nodes. Minimizing the sum of squared drifts at that
268 fixed mean therefore rewards equalizing drift across nodes. In the chosen
269 frequency-cluster model, this favors coupling between unlike-frequency clus-

ters, whereas the diagnostic template rewards coupling within like-frequency clusters. The objective and the scored structure pull in different directions.

For fixed phases, the continuum objective is

$$\sigma(K) = \frac{1}{D} \int \left[\omega(x) + \int K(x, y) \sin(\theta_y - \theta_x) d\mu(y) \right]^2 d\mu(x). \quad (6)$$

The formalization constructs the bounded drift operator A and its adjoint, obtains $\nabla\sigma = (2/D)A^*(AK + \omega)$, and proves descent for the specified gradient flow. This is a squared-drift functional; identifying thermodynamic entropy production would require probability-current information.

The finite control co-evolves noisy phases and symmetric, nonnegative couplings at fixed total resource. The frequency-cluster template is a diagnostic covariance of shared latent signals, not a training input. Across three seeds, second-half order remains between 0.9167 and 0.9222, while the within-cluster to between-cluster coupling ratio falls from 0.9824–0.9948 to 0.3507–0.6135 (Fig. 4). The supplement reports random-update, relabelled-gradient and frozen-kernel controls, including comparisons at overlapping deformation and coherence ranges and against partitions blind to frequency.

The objective reduction is transient: averaging over complete update intervals removes the apparent gain seen immediately after updates. A pre-specified grid of 6 rate–cadence combinations contains no case meeting the declared 5% sustained-reduction criterion while retaining coherence and losing alignment. The finding is coherent adaptation away from the specified

290 structure, with no demonstrated sustained cost reduction.

291 Kernel alignment and overlap compatibility are different observables.
292 This control tests the narrower suggestion that coherent adaptation under
293 this objective acquires the designated environmental structure. It fails for
294 an intelligible reason: the objective rewards a different organization. A
295 thermodynamically specified learning model can include data in its objective
296 (Corberi et al., 2026); explaining agreement about the cup would likewise
297 require a learning rule connected to the relevant content.

298 **4.3 What must be specified and measured**

299 Compatibility enters the construction as a hypothesis about the local states.
300 For deterministic positive coupling, convergence to a potential minimum is
301 proved under an explicit initial basin condition; the proposed bridge E78
302 connects supercritical noisy coherence to relaxation of the specified cover.
303 Neither step explains how initially inconsistent representations become con-
304 sistent. That is the mechanism a biological account must add.

305 A neural test must first identify the domains and objects whose agreement
306 is at issue. Two decoders should estimate a shared content on overlapping
307 domains, with restriction maps specified before inspecting their agreement.
308 Their discrepancy needs a null under decoding noise, and the procedure must
309 distinguish deliberately constructed compatible and incompatible examples.
310 A common prior can make separate decoders agree artificially; region-specific
311 bias can make them disagree artificially. The supplement develops these re-

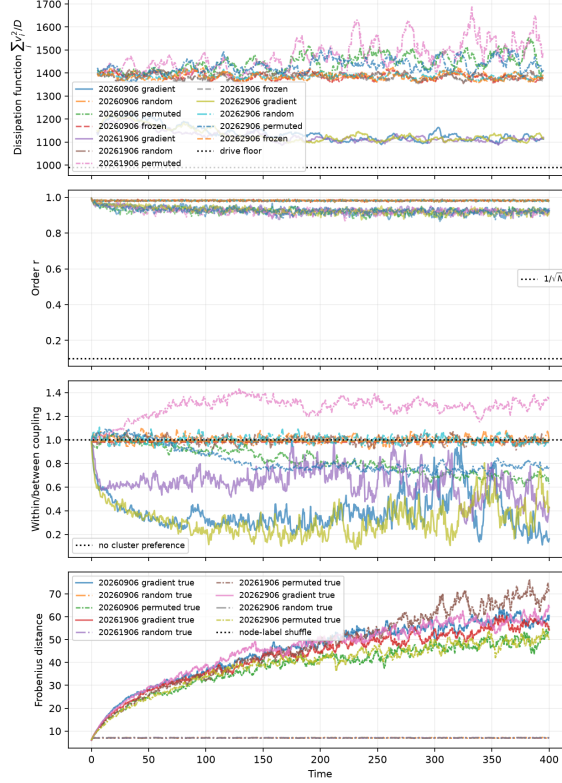


Figure 4: Coherence persists while coupling loses alignment with the frequency-cluster template. Gradient updates (solid), random updates (dash-dotted), relabelled gradient updates (dash-dot-dotted), and a frozen kernel (dashed) share one phase-noise stream per seed. Panels show the immediate post-update squared-drift objective, phase order, within-cluster to between-cluster coupling, and template distances. The gradient arm preserves order while its coupling ratio falls; growth in both distances mainly reflects kernel norm. The objective panel samples the instant after updates; complete-interval measurements find no sustained reduction in the bounded study.

312 quirements. An estimator built from phase agreement would miss the prop-
 313 erty the counterexample separates.

314 Choosing a cover is also part of choosing a candidate subject. Trivial
 315 descriptions can agree while carrying little content, and a suitably chosen
 316 partition can conceal disagreement. The scientifically useful choice must
 317 therefore be constrained independently by what local activity encodes and
 318 which shared quantities can be compared. The uniqueness theorem then
 319 answers a precise question about that choice: whether the supplied local
 320 descriptions determine one global state. It leaves the empirical selection of
 321 domains and descriptions to the model.

322 **5 Self-representation: a state that models it-** 323 **self**

324 **5.1 The reflexive construction**

325 Compatibility yields a global state. Reflexivity adds a relation between that
 326 state and an encoding within it. Equip global sections with a complete
 327 metric. Under the additional restriction-resonance condition, the encoding
 328 of a section s is its restriction to a region U , and the map takes the form

$$F(s) = \text{readout}(s|_U). \tag{7}$$

329 The same region and readout are used across the model’s state space. If F
330 contracts and fixes the section s_* supplied by the cover, Banach’s theorem
331 makes s_* its unique fixed point (Banach, 1922). The formal bridge E89 sup-
332 plies the contraction law and $F(s_*) = s_*$ for the boundary’s encoding–readout
333 composite. Identifying that encoding with restriction is a separate resonance
334 condition; E89 does not supply it. The supercritical order parameter pro-
335 vides an inequality used by the specified contraction bound; constructing the
336 encoding and readout remains a separate obligation.

337 The equality ties the reconstruction to the state formed from the local
338 descriptions. In the cup example, the candidate global state would include
339 the represented scene and the perceiver’s relation to it. A local encoding
340 would have to retain enough information for the specified readout to recover
341 that modelled state. An encoding of cup position alone would be insufficient
342 if the readout could not recover the represented hand and viewpoint. The
343 term *predictive* names this reconstruction map; a temporal neural prediction
344 requires a dynamical interpretation.

345 **5.2 Why connect these properties to unity and a min-** 346 **imal self?**

347 We identify the section glued from compatible local states with the unity
348 of a conscious episode, and its accurate internal reconstruction with a mini-
349 mal self. Two features of the episode motivate this interpretation: different

350 contents belong to one situation, and that situation is experienced from a
351 perspective.

352 The unity proposal starts from a distinction between having several con-
353 tents and having several experiences. Seeing the cup’s surface and feeling
354 the reaching hand contribute different contents to one situation. Compat-
355 ibility expresses their agreement about what they share while preserving
356 those differences. An episode can also contain uncertainty or conflicting
357 judgments. A model might represent the cup’s apparent position separately
358 from a judged position and consistently describe the discrepancy. Deciding
359 whether two estimates concern the same quantity or distinct aspects of ex-
360 perience is therefore part of defining the content model. A choice made only
361 to ensure agreement would give the unity proposal no empirical force.

362 The motivation for a minimal self is that the cup is encountered from a
363 location and in relation to a body that can reach it. The proposed internal
364 representation concerns the perceiver’s condition in that encounter: where
365 the perceiver is situated, what is being represented and how the body relates
366 to it. Such a representation could be implicit in the organization of activity;
367 it need not describe a person verbally or carry an autobiographical narrative.
368 Requiring it to reconstruct the same global state links the perspective to the
369 episode whose contents it organizes.

370 The fixed-point condition requires accuracy about the very state contain-
371 ing the encoding. Reconstruction alone is insufficient for a perspective: a
372 recording can be reconstructed without being used by a system to represent

373 its own condition. The proposed interpretation therefore needs evidence that
 374 the encoding helps the system distinguish changes in itself from changes in
 375 its surroundings. Changing the encoded hand or viewpoint while holding
 376 the external scene fixed should alter the system’s estimates of its relation
 377 to the cup as the readout predicts. This functional criterion goes beyond
 378 $F(s_*) = s_*$; whether the represented perspective is experienced remains a
 379 further question.

380 Accuracy concerns the variables represented by s_* , rather than every mi-
 381 croscopic physical fact. For a fixed map F with Lipschitz factor $q < 1$ and
 382 fixed point s_* , approximate reconstruction has a quantitative bound:

$$d(s, F(s)) \leq \varepsilon \implies d(s, s_*) \leq \frac{\varepsilon}{1 - q}. \quad (8)$$

383 At the stipulated rate $q = e^{-(K-2D)\tau/2}$, this radius grows as $2\varepsilon/((K-2D)\tau)$
 384 near threshold, for fixed positive ε and τ . The bound weakens while the
 385 exact fixed point remains unique. The supplement gives explicit inequalities
 386 and composes reconstruction error with displacement of the state. A corti-
 387 cal realization must specify the represented variables, their metric and the
 388 required fidelity; changing readouts require a further dynamical model.

389 We propose reflexivity as necessary because even an unreflective episode
 390 has a perspective: the cup appears somewhere relative to the perceiver,
 391 whether or not the perceiver attends to themselves. Requiring internal self-
 392 encoding on that basis is the philosophical hypothesis. Gluing alone does

393 not require it; the dissociations below test the two commitments separately.

394 **5.3 Which cover and which self-model?**

395 Both uniqueness results are conditional on their objects. Gluing gives one
396 global extension of specified compatible local data; Banach gives one fixed
397 point of a specified contracting map. These solve ambiguity within a chosen
398 construction. They do not choose between different covers, different encod-
399 ings or different candidate subjects. Integrated information theory makes
400 exclusion an explicit part of its account (Tononi et al., 2016); the present
401 uniqueness results do not settle that selection problem.

402 The relevant cover and map must be tied to neural organization by evi-
403 dence independent of the desired conclusion. Local descriptions should pre-
404 dict the contents attributed to their domains, and their restrictions should
405 preserve meaning on shared domains. The encoding and readout should be
406 specified before testing them, with their sensitivity to changes in the pro-
407 posed state measured over a declared range.

408 A constant readout returning one predetermined state contracts and has a
409 fixed point, even when it discards an encoding supplied by restriction. Such a
410 construction fails the sensitivity requirement above. This is why selecting the
411 map requires evidence about its functional role in addition to mathematical
412 existence.

413 5.4 Dissociations that would bear on the identification

414 A test of the full commitment must measure its clauses separately: effective
415 coupling relative to diffusion, overlap compatibility, and the accuracy of the
416 proposed self-encoding, alongside an independent endpoint for experience.
417 A case meeting all three physical and representational conditions with evi-
418 dence against a unified conscious episode would challenge their sufficiency.
419 Unified experience with demonstrably incompatible local descriptions would
420 challenge the unity identification. Unified experience with compatibility but
421 without the specified self-encoding would challenge the reflexive requirement.

422 Anesthetic emergence, dreaming and dissociative states are candidate set-
423 tings because the framework asks whether coordination, unified content and
424 self-representation can separate. Interpreting such evidence requires a de-
425 clared relation between the experience endpoint and report; absence of report
426 alone cannot define absence of an episode. The compatibility observable in
427 the supplement is a specification, and the corresponding neural self-encoding
428 has yet to be identified. These are the main obstacles to testing the inter-
429 pretive commitment.

430 The proposal concerns which state is conscious. It supplies no explanation
431 of why satisfying these relations should be accompanied by experience. Nor
432 do its gluing and fixed-point conditions select an electromagnetic substrate:
433 the cortical field commitment resides in the physical identification and cali-
434 bration. Their multiple realizability leaves the status of other biological and
435 artificial implementations open.

436 6 Thermodynamic constraints and the con- 437 ditional composition

438 The broader composition constrains one proposed physical implementation
439 through finite-register capacity $H(p) \leq \log |X|$, erasure heat under a specified
440 system–bath model (Landauer, 1961; Norton, 2005; Shenker, 2000), and a
441 separate field-boundary result for disconnected vacuum components (Kibble,
442 1976). The supplement develops these results and their witnesses. The phase
443 threshold and representational argument stand independently of a derivation
444 from finite capacity.

445 The constraint most directly relevant to representation concerns predic-
446 tive memory. For a state X_t driven by an environmental signal S_t ,

$$W_{\text{diss}} \geq k_B T [I(X_t; S_t) - I(X_t; S_{t+1})]. \quad (9)$$

447 Under the no-back-action Markov structure $X_t \rightarrow S_t \rightarrow S_{t+1}$, the bracket is
448 nonnegative (Still et al., 2012). The bound limits nonpredictive memory at a
449 given dissipation budget; zero retained information satisfies it too. Relating
450 it to Landauer heat requires that memory waste be bounded by erased en-
451 tropy at the same thermal scale and heat budget. These distinct information
452 constraints (Rosas et al., 2026) leave content and learning dynamics to be
453 specified.

454 An acting agent changes its future inputs, so the passive assumption

455 requires justification for the variables and interval considered. A feedback
 456 model specifies a policy $A_t \sim \pi(\cdot | X_t)$, world dynamics $S_{t+1} \sim K(\cdot | S_t, A_t)$,
 457 an observation channel and a memory update. For finite states, the exact
 458 information balance is

$$\begin{aligned} I(X_t; S_t) - I(X_t; S_{t+1}) &= I(X_t; S_t | S_{t+1}) \\ &\quad - I(X_t; S_{t+1} | S_t). \end{aligned} \tag{10}$$

459 Both conditional terms are nonnegative. The last term vanishes for the
 460 passive model; with feedback, the signed difference can be negative because
 461 actions create correlations with the future. Its causal interpretation requires
 462 an environmental state sufficient to account for the relevant dynamics.

463 The finite thermodynamic extension instead bounds joint entropy reduc-
 464 tion. For an elementary update that holds the controller fixed, strictly
 465 positive forward and reverse path probabilities give $\Sigma = D_{KL}(P_F || P_R) =$
 466 $\Delta H(X, S) + \langle Q \rangle / (k_B T) \geq 0$ under local detailed balance, with Q positive
 467 into the reservoir (Horowitz and Esposito, 2014; Ito and Sagawa, 2013). This
 468 supplies the thermodynamic constraint for the active branch below. A proved
 469 two-substep extension composes actuation with a memory update initial-
 470 ized from its coordinate-swapped final law. It identifies the output with the
 471 perception-action channels and cancels the same intermediate entropy and
 472 expected energy. At a common thermal scale, the whole update's joint en-
 473 tropy decrease is bounded by its total heat budget divided by $k_B T$, without
 474 assuming stationarity. The supplement gives the balances and reciprocal

475 two-bit witnesses with explicit interaction energy, heat and work. Action
476 and observation alias the existing states; additional physical registers require
477 further accounting.

478 An explicit objective also permits finite policy selection. For a supplied
479 terminal reward and a finite set of deterministic policies, a nonempty set sat-
480 isfying the complete update’s heat budget has a reward maximizer. A two-bit
481 lamp task has a unique feasible optimum that strictly improves on a feasi-
482 ble constant-action baseline; a stronger policy exceeds the budget. Reward,
483 heat and work are evaluated on each policy’s actual composed process, with
484 common initial law, channels, energy and reservoir convention. The base-
485 line meets the heat budget while missing the task target, so the objective
486 supplies content that thermodynamics alone does not. This result compares
487 fixed policies for one update.

488 A separate finite learning model specifies how a policy register adapts. Its
489 bit selects between the two feasible lamp policies, and its energy decreases
490 with their known expected reward. An autonomous thermal channel changes
491 the register, carrying each final distribution into the next update. From the
492 specified initial preparation, Lean proves strictly increasing expected task
493 reward and geometric convergence to a stationary mixture. Transition heat
494 obeys local detailed balance, and cumulative update heat equals the regis-
495 ter’s finite energy loss at zero external work. The mixture remains below
496 the optimal policy; individual updates can lower reward, and a different
497 initial preparation lowers expected reward. This is adaptation toward an

498 encoded objective. Learning unknown rewards from observations, preparing
 499 the register and implementing its readout in an actuator require additional
 500 accounting. Task reward and task heat are prospective expectations under
 501 the prepared lamp law; the learning clock executes no task episodes and
 502 supplies no spatial coupling convergence or cortical identification.

503 6.1 What the composition establishes

504 Table 1 lists the connecting hypotheses of two branches. The theorem **chain**
 505 uses passive predictive memory; **chain_active** uses a specified finite feedback
 506 process. Each accepts eight implications and shares the passage from coarse-
 507 graining to the phase model and its proposed representational role. In both,
 508 the finite register and the field boundary are independent starting objects.

509 For the active branch, the controller has the register’s state type and the
 510 fourth node records

$$\theta > 0, \quad \theta[H(p) - H(p')] \leq \langle Q \rangle \leq B, \quad (11)$$

511 where p, p' and $\langle Q \rangle$ belong to that same controlled substep. E34Active sup-
 512 plies positive path probabilities, local detailed balance at the register’s ther-
 513 mal scale θ , and the allocation $\langle Q \rangle \leq B$, with B equal to the register’s
 514 heat. Landauer’s lower bound does not imply this upper-budget allocation
 515 or identify erasure with actuation. E45Active requires the bound for these
 516 specified data to place the coupling energies in the named convergent regime.

517 A coupling dynamics establishing that implication is an additional modelling
518 requirement.

519 Two structures also enter both theorems: a thermodynamic cover, in-
520 cluding overlap compatibility, and a reflexive boundary specifying the local
521 encoding and readout. Given these and the connecting hypotheses, both
522 conclude **UnifiedSelf**: the section glued on that cover is the unique fixed
523 point of self-prediction. Restriction resonance, needed for the interpretation
524 in Eq. (7), is a separate condition. The result binds the fixed point to the
525 specified local states; compatibility and accuracy at that state are premises,
526 while uniqueness follows from the construction.

527 The joint witnesses combine a double well, a register, a changing uniform-
528 grid mesh and a three-site cortex; the active branch includes the noisy two-bit
529 actuator. They establish that the hypotheses can hold together in distinct
530 toy components. Mesh convergence is supplied independently of the thermo-
531 dynamic bounds, and no common biological mechanism is established. The
532 active witness rejects a zero heat budget and an incompatible energy limit.
533 Counterexamples also expose the role of overlap agreement. The develop-
534 ment declares no additional Lean axioms beyond the foundational axioms
535 used by its libraries; physical premises are explicit hypotheses and model
536 fields.

Connection	Status	Required assumption
E12	Independent physical premise	A finite-capacity system realizes a field that leaves its vacuum manifold. Capacity and the boundary are independent roots.
E23	Bridge assumption	The boundary is associated with a finite register whose update leaves states unreachable.
E34 / E34Active	Bridge assumption	Passive: a predictive structure with the register’s thermal scale and heat budget, and nonzero wasted memory. Active: positivity, local detailed balance and the register’s heat allocation for the named elementary process. Complete-update accounting, finite control and policy-register adaptation do not derive this common resource allocation.
E45 / E45Active	Modelling assumption	The branch’s bound, on predictive memory or the named feedback process, places coupling energies in the specified convergent coarse-graining regime.
E56	Physical commitment	A specified physical field realizes the limiting coupling and satisfies the kernel conditions.
E67	Physical commitment	That field operates above the synchronization threshold.
E78	Bridge assumption	Coherence drives a supplied cover to its relaxed equilibrium at coupling at least the named mean-field value.
E89	Modelling assumption	A map mediated by local self-encoding fixes the section glued on that cover and has the Lipschitz factor $q < 1$ used in the exact and approximate-state results.

Table 1: Connecting hypotheses of the passive and active compositions, eight per branch. They share E12, E23 and E56–E89. The cover’s compatibility is supplied as a structure field. Identifying the boundary’s encoding with restriction requires a separate resonance condition. The mathematical objects and node-level results are mapped in Supplemental Material, Table S1.

537 7 Discussion

538 The strongest results concern what a coherence measurement leaves unre-
539 solved. Different recovery mechanisms share stationary distributions; equal
540 phases permit incompatible local descriptions; and the tested plasticity ob-
541 jective sustains coherence while moving away from its diagnostic structure.
542 These findings make the coupling hypothesis more demanding to test and
543 prevent a successful synchronization model from doing unexplained repre-
544 sentational work.

545 The proposed account identifies a compatible, internally represented state
546 with a conscious episode. Its main explanatory gap is the mechanism that
547 would maintain those relations: compatibility and accuracy at the glued state
548 enter the construction as premises. Their mathematical consequences leave
549 open why cortical activity should acquire or preserve them.

550 The most informative next step for that connection is a test of whether
551 coordination helps maintain independently measured content agreement. Af-
552 ter validating separate decoders and their overlap restrictions, a controlled
553 change in coupling or coordination could test whether agreement deterio-
554 rates in the predicted way, with controls for changes in sensory information
555 and decoder quality. Such a result would bear directly on the proposed role
556 of coherence. In parallel, the awakening hypothesis needs its own calibrated
557 geometry–coupling relation and resolved phase dynamics to distinguish a cou-
558 pling increase from diffusion change or common drive. A favorable recovery

559 trace alone would leave the representational connection open.

560 Testing the self clause requires perturbing a candidate encoding and com-
561 paring the resulting changes in represented scene, body and viewpoint with
562 the readout’s predictions. Establishing that functional role would make dis-
563 sociations from unified experience informative. The exact fixed-state ideal-
564 ization must also be assessed against the fidelity and time dependence of the
565 biological process.

566 Integrated-information, global-workspace and predictive-processing ac-
567 counts organize unity, access and modelling differently (Tononi et al., 2016;
568 Baars, 1997; Dehaene et al., 2014; Friston, 2010). The present proposal
569 contributes a candidate interaction mechanism and explicit relations to test
570 between local content and internal self-encoding. Its explanatory value will
571 turn on whether those relations can be identified independently and shown
572 to matter for experience. For the cup example, that means explaining how
573 the system maintains one represented situation and how the perceiver’s place
574 within it contributes to the episode.

575 **Data and code availability**

576 The Lean development, simulation scripts, tests and saved numerical sum-
577 maries are maintained in the manuscript source repository. Publication sim-
578 ulation values are regenerated from saved summaries without rerunning pro-
579 duction sweeps. The exploratory EEG analysis uses OpenNeuro dataset

ds005620 (Bajwa et al., 2025); preprocessing and selection are described
in Supplemental Material. The source repository contains the reproducible
analysis and does not redistribute the raw recordings.

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Use of AI tools

AI assistance to this work was substantial and is not separable by task. Several coding assistants and chat interfaces were used in an interleaved way throughout: Google’s Antigravity CLI (Gemini 3.1, Claude 4.6 Opus, Claude 4.6 Sonnet), OpenAI’s Codex CLI (Astra, Terra, Luna, Sol), Claude Code (Claude Opus 5), Gemini Pro through its web interface, and Fable (Claude Opus 4) through OpenRouter, accessed with the Hermes CLI. Between them they contributed to the Lean development, the simulation and analysis code, and the drafting and revision of both documents. Because the tools were used in combination and in alternation, no section, proof, figure or result is attributable to any one of them, and none is claimed to be. The author is responsible for all content, including everything a tool produced. Machine checking establishes the stated Lean results under their hypotheses; it does

not validate biological interpretations or replace scientific review of generated
code and prose.

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